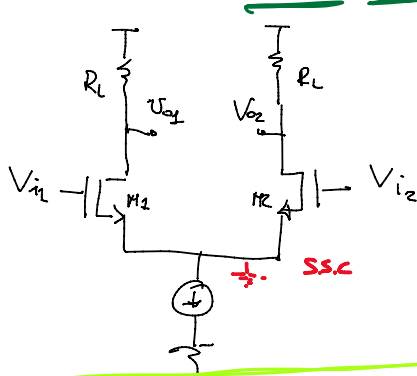


Lecture-13

Revanth Reddy
Pannala, EBIT
Unibo

Evolutions of the differential pair
lunedì 22 marzo 2021 14:58

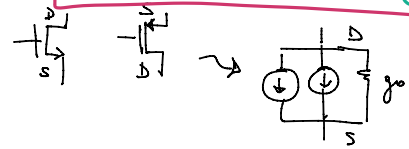
NMOS Diff. Pair



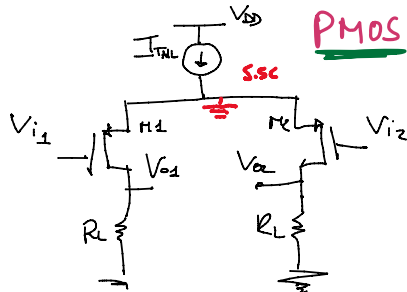
$$[(V_{i1} - V_{i2})]$$

$$\frac{V_{i1} + V_{i2}}{2} = 0$$

$$V_{o1} = - \frac{g_{m1}}{g_{o1} + G_L} V_{i1} ; V_{o2} = - \frac{g_{m2}}{g_{o2} + G_L} V_{i2}$$

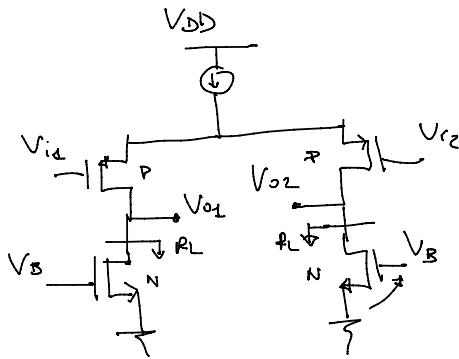


PMOS Diff. pair

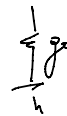


$$V_{o1} = - \frac{g_{m1}}{g_{o1} + G_L} V_{i1} ; V_{o2} = - \frac{g_{m2}}{g_{o2} + G_L} V_{i2}$$

$$(V_{o1} - V_{o2}) = - \frac{g_m}{g_o + G_L} (V_{i1} - V_{i2})$$

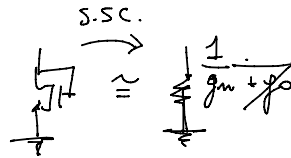
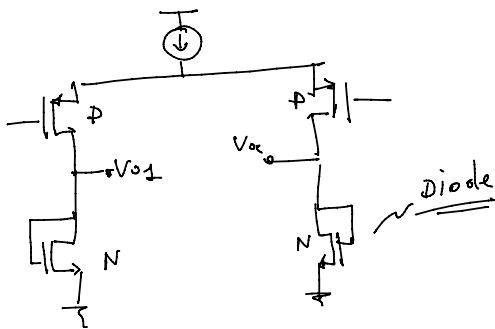


$$\rightarrow \text{S.S.W } V_{ps} = 0$$



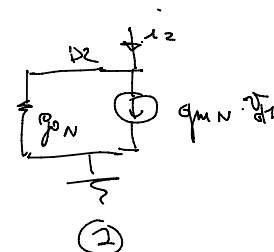
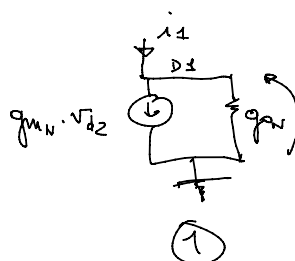
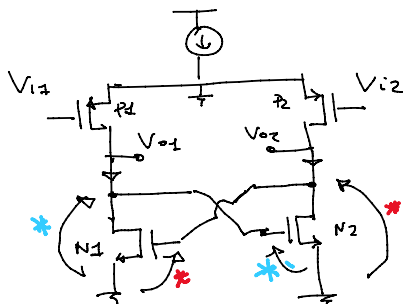
$$R_L = \frac{1}{g_o} ; G_L = g_o$$

$$\frac{(V_{o1} - V_{o2})}{(V_{i1} - V_{i2})} = - \frac{g_{mP}}{g_{oP} + g_{oN}}$$



$$G_L = g_{mN}$$

$$\frac{(V_{o1} - V_{o2})}{(V_{i1} - V_{i2})} = - \frac{g_{mP}}{g_{oP} + g_{mN}}$$



$$i_1 = g_{mN} \cdot V_{d2} + g_{oN} \cdot V_{o1}$$

$$i_2 = g_{mN} \cdot V_{d1} + g_{oN} \cdot V_{o2}$$

$$V_{d1} = V_{o1} \quad V_{o1} = V_{d1} - V_{d2}$$

$$i_1 = g_{mN} \cdot v_{d2} + g_{oN} \cdot v_{d1}$$

$$i_2 = g_{mN} \cdot v_{d1} + g_{oN} \cdot v_{d2}$$

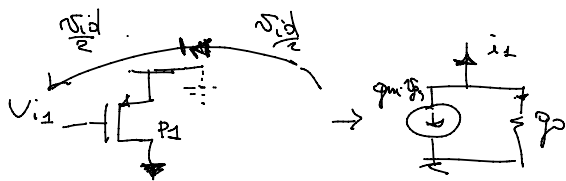
$$(i_1 - i_2) = g_{mN}(v_{d2} - v_{d1}) + g_{oN}(v_{d1} - v_{d2})$$

$$(i_1 - i_2) = (-g_{mN} + g_{oN})(v_{d1} - v_{d2}) = (-g_{mN} + \beta)(v_{od})$$

$$v_{d1} = v_{o1}$$

$$v_{od} = v_{o1} - v_{o2}$$

$$v_{d2} = v_{o2}$$



$$v_{cm} = 0$$

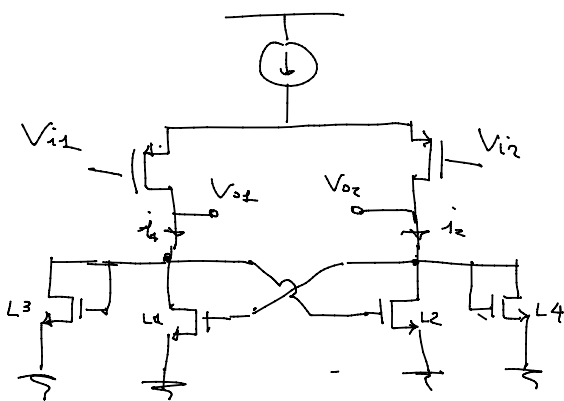
$$v_{id} = \frac{v_{id}}{2}$$

$$v_{i1} = -\frac{v_{id}}{2}$$

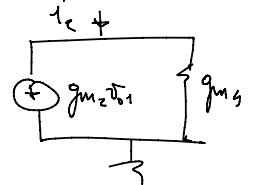
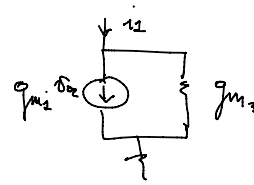
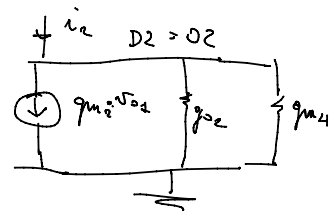
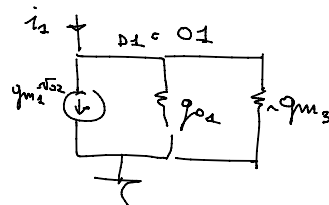
$$i_1 = -g_{mP} \frac{v_{id}}{2} ; i_2 = g_{mP} \frac{v_{id}}{2}$$

$$-g_{mP} \frac{v_{id}}{2} - g_{mP} \frac{v_{id}}{2} = (-g_{mN} + \beta) v_{od}$$

$$v_{id}(-g_{mP}) = (-g_{mN} + \beta) v_{od} \Rightarrow \frac{v_{od}}{v_{id}} = \frac{g_{mP}}{g_{mN} + \beta} \approx \frac{g_{mP}}{g_{mN}}$$



S.S.C



$$\frac{W}{L}_1 = \frac{W}{L}_2 \Rightarrow g_{m1} = \beta g_{m2}$$

$$\frac{W}{L}_3 = \frac{W}{L}_4 \Rightarrow g_{m4} = \beta g_{m3}$$

$$(i_1 - i_2) = g_{m1}(v_{d2} - v_{d1}) + g_{m3}(v_{d1} - v_{d2})$$

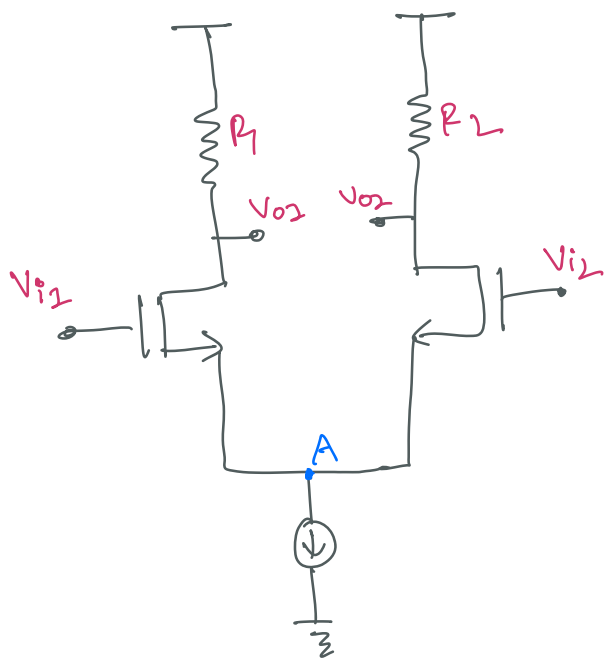
$$(i_1 - i_2) = (-g_{m1} + \beta g_{m3})(v_{d1} - v_{d2}) = (-g_{m1} + \beta g_{m3}) v_{od}$$

$$\left[\frac{v_{od}}{v_{id}} = -\frac{g_{mP}}{g_{m3} - g_{m1}} \right]$$

⇐

→ Today we will further elaborate the concept of Differential pair and discuss possible improvements to different contexts in which Diff. pair will be used.

First of all the basic Differential pair with balanced couple of Branches Fed by common current source.



→ Here we are interested in the difference in variations of both inputs

$$V_{i1} \text{ \& \; } V_{i2}$$

→ we want to amplify $(V_{i1} - V_{i2})$ and restrict amplification of $\frac{V_{i1} + V_{i2}}{2}$

→ we went through the analysis of Diff. pair and when we are focusing on pure Differential Analysis and we assume

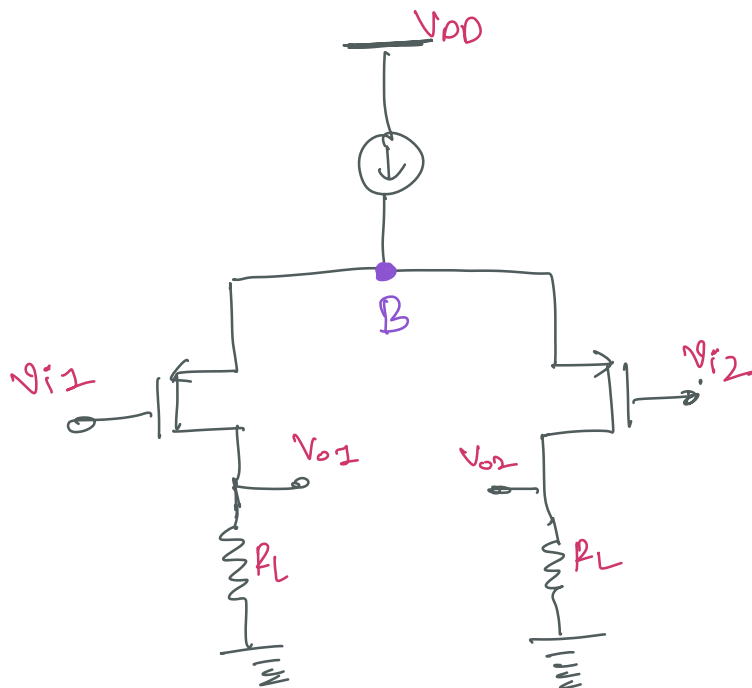
$$V_c = \frac{V_{i1} + V_{i2}}{2} = 0$$

→ Then for the small signal circuit at node A we have a Virtual Ground.

so basically we can analyze two branches independently. and it seems like two common source stages operating independently.

i.e.
$$V_{o1} = \frac{-g_{m1} V_{i1}}{g_{o1} + G_L}, \quad V_{o2} = \frac{-g_{m2} V_{i2}}{g_{o2} + G_L}$$

Differential Pair using PMOS configuration

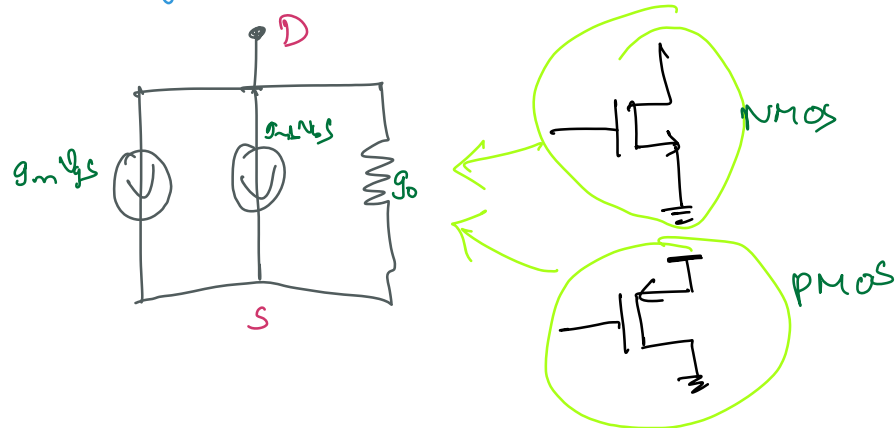


→ Once again if we can neglect common mode variations i.e.
$$V_c = \frac{V_{i1} + V_{i2}}{2} = 0$$

Then in SSM we have a virtual GND at Node B.

and this means we have two separate CS Configurations as in NMOS Diff pair. and we can reapply same machinery we have had before.

→ we know that NMOS & PMOS have same Small signal Transistor Models.



If SSM of NMOS & PMOS are same we can conclude their gains are also same.

$$\therefore v_{o2} = \frac{-g_{m2} v_{i2}}{g_{o2} + G_L} \quad v_{o2} = \frac{-g_{m2} v_{i2}}{g_{o2} + G_L}$$

→ Total equivalent for NMOS & PMOS.

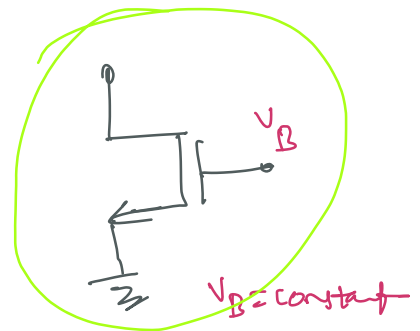
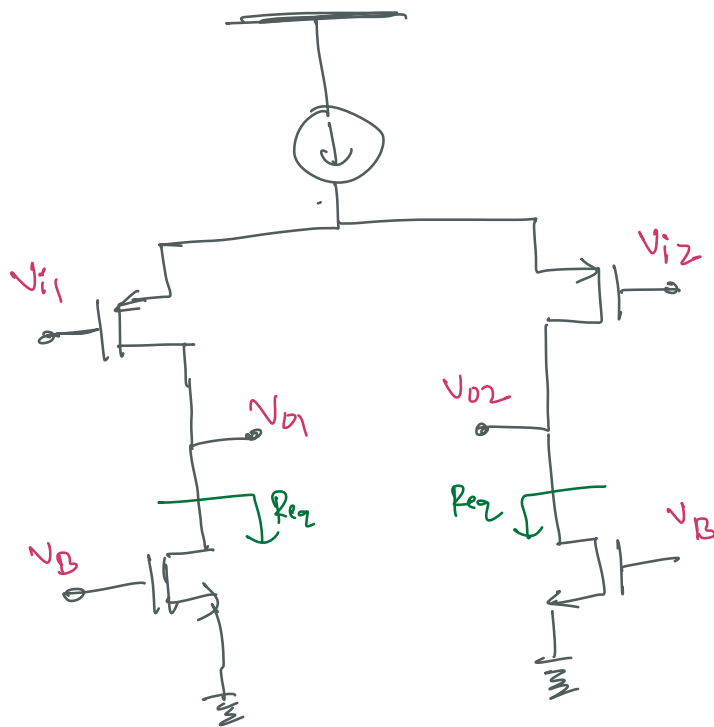
ofcourse Transconductance of NMOS can be Different from Transconductance of PMOS. because ...

because Mobilities $\rightarrow$ changing and they vary.

$$V_{O1} - V_{O2} = \frac{-g_m}{g_{O1} + g_{O2}} (V_{i1} - V_{i2})$$

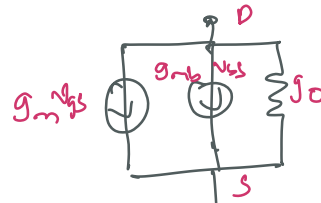
If Transistors are equally size and have same Quiescent point & Same loading Resistors

$\rightarrow$ In the practice Diff. pair circuits in place of Resistors we use other Transistor for implementing Resistors -

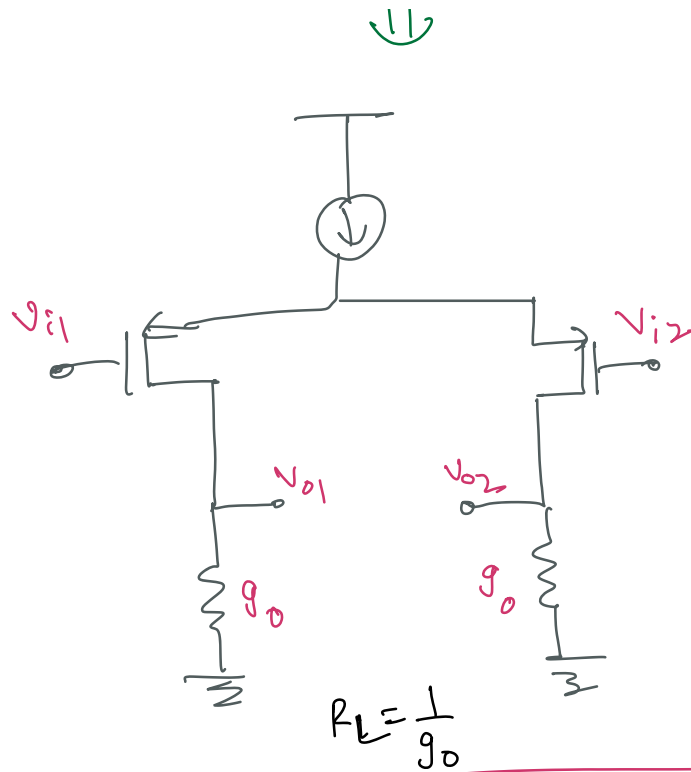


R_{eq} = Equivalent Resistive effect of Transistor as shown in the figure

SSM of NMOS with $V_B = \text{constant}$



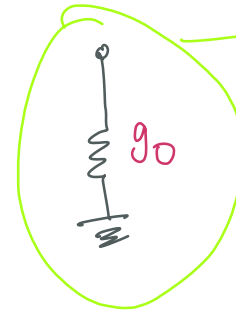
||



$\therefore V_B = \text{constant} = \text{Gate voltage}$

$$V_{gs} = 0$$

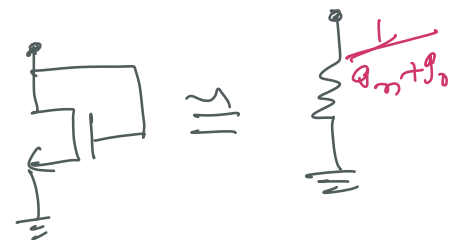
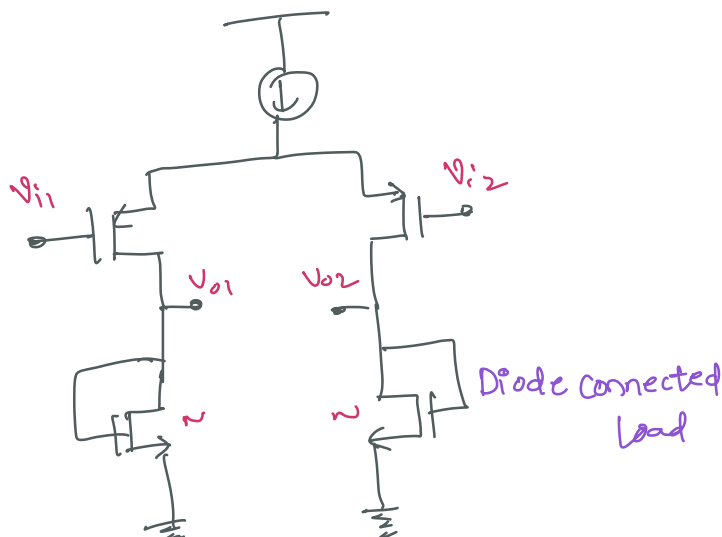
$$V_{bs} = 0$$



Only SSM of NMOS as load.

$$\therefore \frac{V_{o1} - V_{o2}}{V_{i1} - V_{i2}} = \frac{-g_{mp}}{g_{op} + g_{on}} \quad \text{--- (1)}$$

An other possibility of implementing loading Resistors



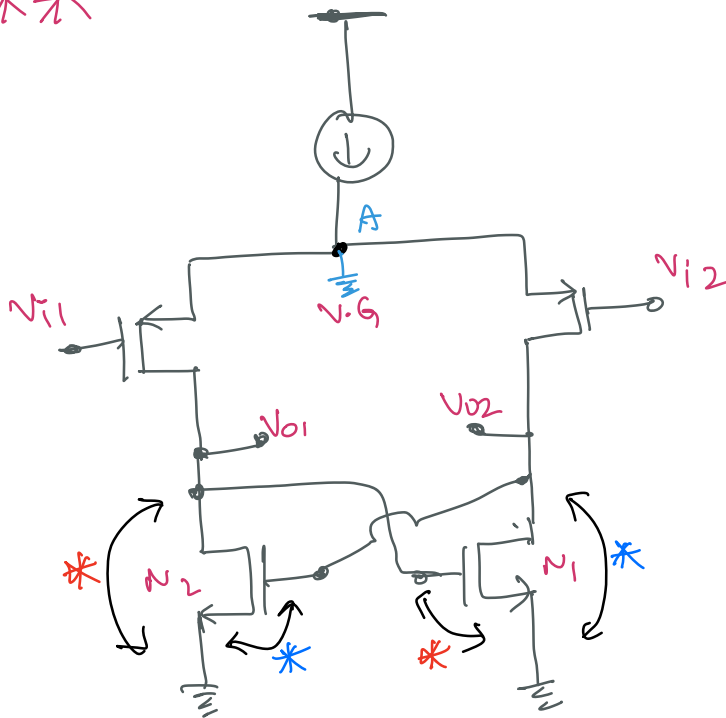
$$\frac{V_{o1} - V_{o2}}{V_{i1} - V_{i2}} = \frac{-g_{mp}}{g_{op} + g_{mn}} \quad \text{--- (2)}$$

Q Among ① & ② which has higher Gain?

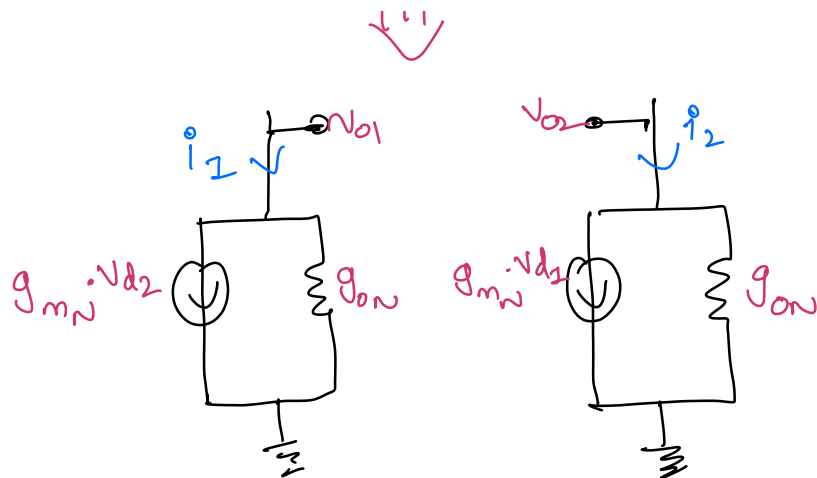
① has more gain because $g_{o_n} \ll g_{m_n}$

Even though ② may have lower gain, it also has some utility.

සමස්ත පද්ධතිය
සැලකීම
EBIT, Unibo



SSM for N_2 & N_1



→ Now let's derive relation b/w i_1 & i_2 and the SSN of v_1 & v_2

$$i_1 = g_{mn} v_{d2} + g_{on} v_{d1}$$

$$i_2 = g_{mn} v_{d1} + g_{on} v_{d2}$$

$$\Rightarrow i_1 - i_2 = g_{mn} (v_{d2} - v_{d1}) + g_{on} (v_{d1} - v_{d2})$$

$$i_1 - i_2 = (-g_{mn} + g_{on}) (v_{d1} - v_{d2})$$

Basically, $v_{d1} = v_{o1}$

$$v_{d2} = v_{o2}$$

$$v_{od} = v_{o1} - v_{o2}$$

$$\Rightarrow i_1 - i_2 = (-g_{mn} + g_{on}) v_{od}$$

we have virtual ground at node (A)

$$i_2 = \frac{g_{mp} V_{id}}{2}$$

To sum up what is the final expression

$$i_1 - i_2 = (-g_{mn} + g_{ow}) V_{od}$$

$$\Rightarrow -g_{mp} \frac{V_{id}}{2} - g_{mp} \frac{V_{id}}{2} = (-g_{mn} + g_{ow}) V_{od}$$

$$\Rightarrow -V_{id} g_{mp} = (-g_{mn} + g_{ow}) V_{od}$$

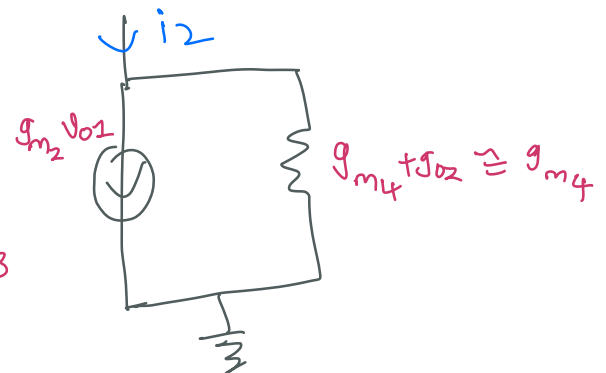
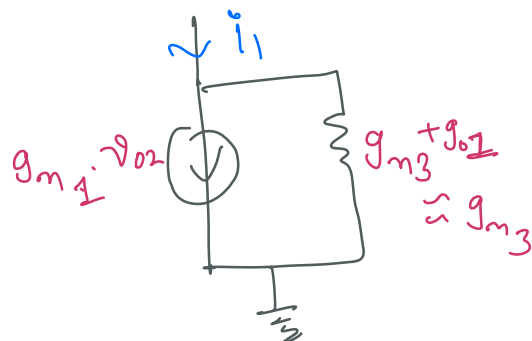
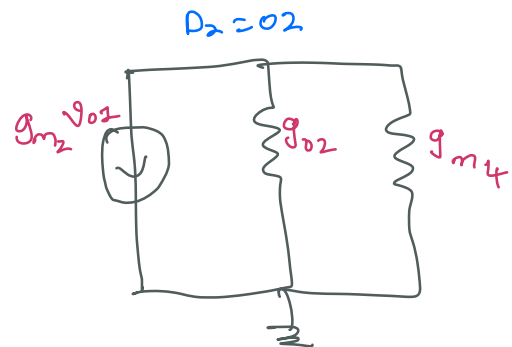
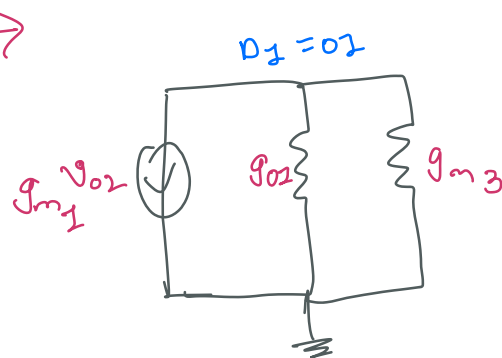
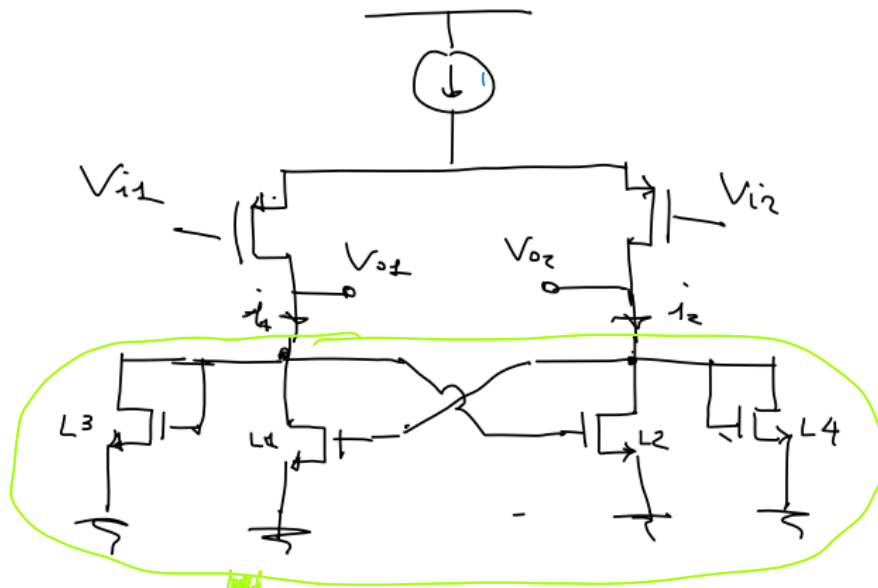
$$\Rightarrow \frac{V_{od}}{V_{id}} = \frac{-g_{mp}}{-g_{mn} + g_{ow}}$$

$$g_{mn} \gg g_{ow}$$

$$\frac{V_{od}}{V_{id}} = \frac{-g_{mp}}{-g_{mn}} = \frac{g_{mp}}{g_{mn}}$$

Here there is no -ve sign
as we have seen in individual
Diode connected loading case -

పాస్‌వోల్డ్‌ రెగ్యులేటర్



→ let suppose all the Transistors are equally sized.

$$\left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2 \Rightarrow g_{m1} = g_{m2}$$

$$\left(\frac{W}{L}\right)_3 = \left(\frac{W}{L}\right)_4 \Rightarrow g_{m3} = g_{m4}$$

$$i_1 - i_2 = g_{m1}(v_{o2} - v_{o1}) + g_{m3}(v_{o1} - v_{o2})$$

$$\Rightarrow \underbrace{i_1 - i_2}_{\downarrow} = (-g_{m1} + g_{m3}) \underbrace{(v_{o1} - v_{o2})}_{V_{od}}$$

after some algebra

$$\frac{V_{od}}{V_{id}} = \frac{-g_{mp}}{g_{m3} - g_{m1}}$$

↓
The closer to g_{m3} to g_{m1}

∴ By selecting g_{m3} almost ^{equal} to g_{m1} we can achieve High Gain.

Differential Amplifier slides



Voltage differential amplifier with $g_m r_{DS}$ gain

$A_v = g_{m1}(r_{DS1} // r_{DS2})$

Willy Sansen 10.05.0327

58

Differential diode-connected MOSTs

$\frac{2}{g_m}$

$-\frac{2}{g_m}$

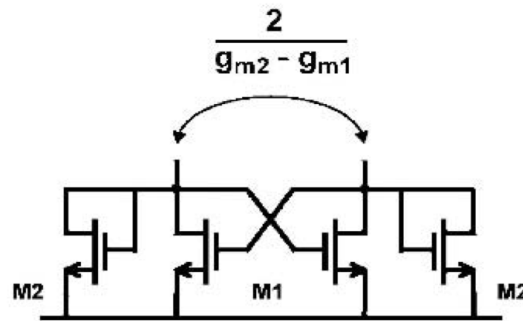
Diode connected loads

Willy Sansen 10.05.0330

59



Differential diode-connected MOSTs

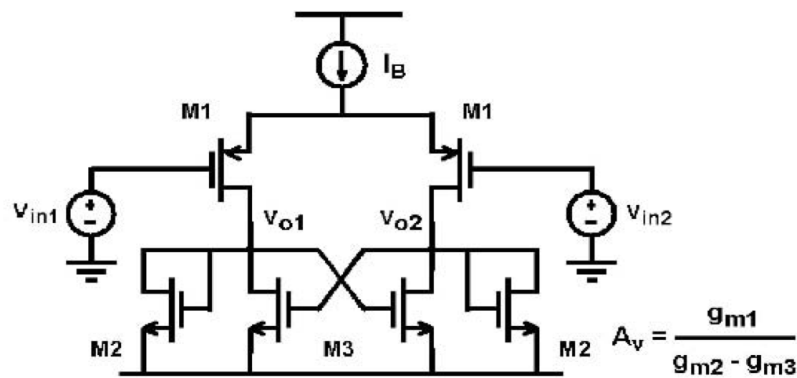


Gain $\rightarrow$ Values close to ∞ ! Infinity

60

Revathi Peddy Pannala
EBIT, Unibo

High gain because of current cancellation



61

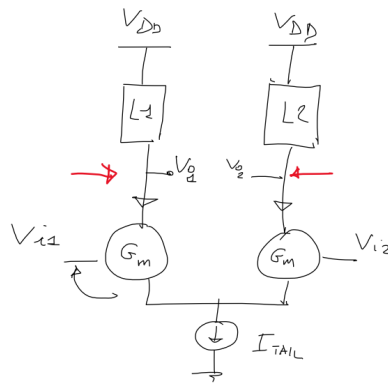
$\rightarrow$ we know that Cascode Configuration has indeed

High o/p Resistance.

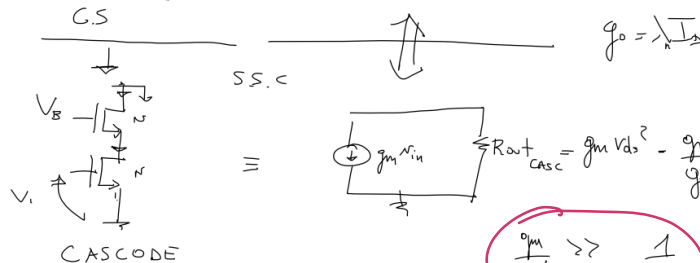
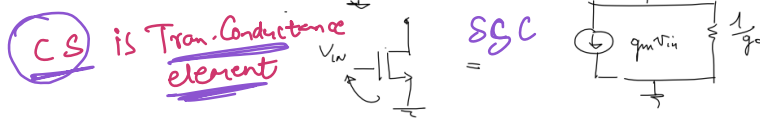
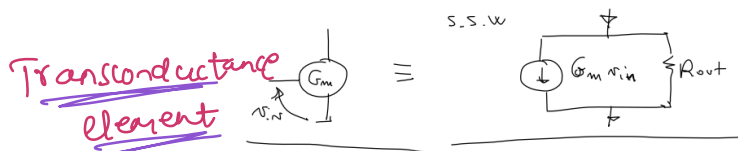
Can we use cascode configuration in the Differential Amplifier?

YES

General scheme of Differential pair



*



$$g_m r_{ds}^2 = \frac{g_m}{g_o^2}$$

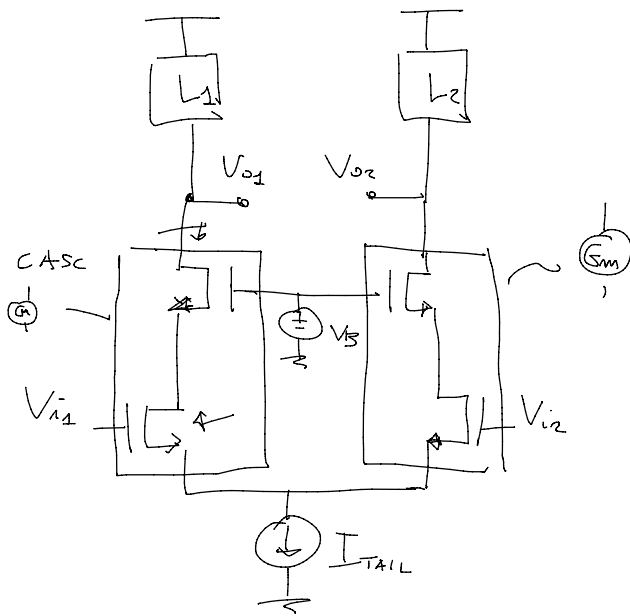
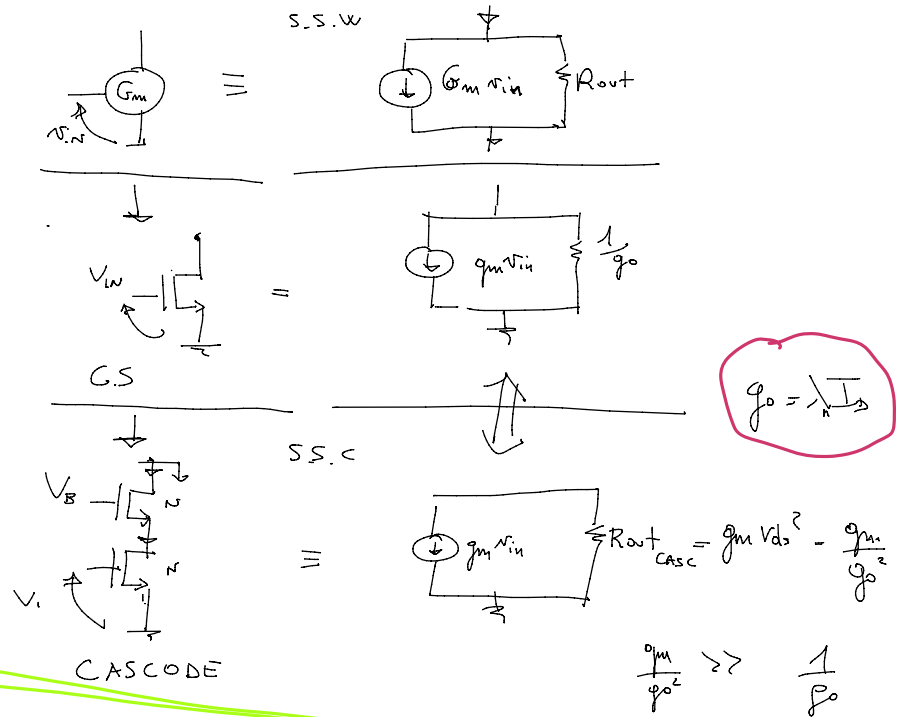
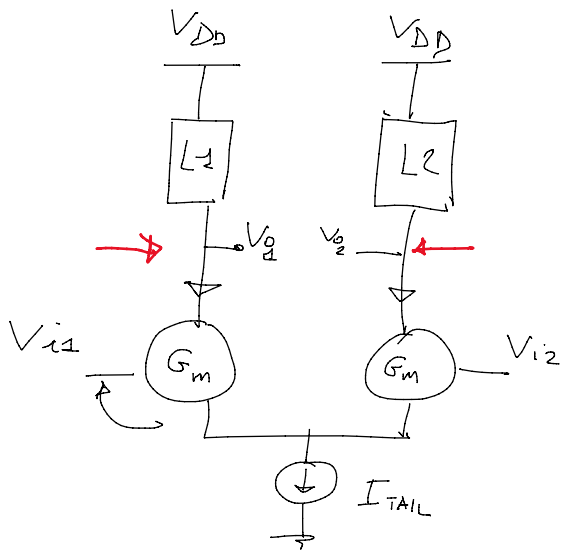
$$\frac{g_m}{g_o^2} \gg \frac{1}{g_o}$$

Cascode is
Tran-Conductance
element

Because o/p
Resistance of

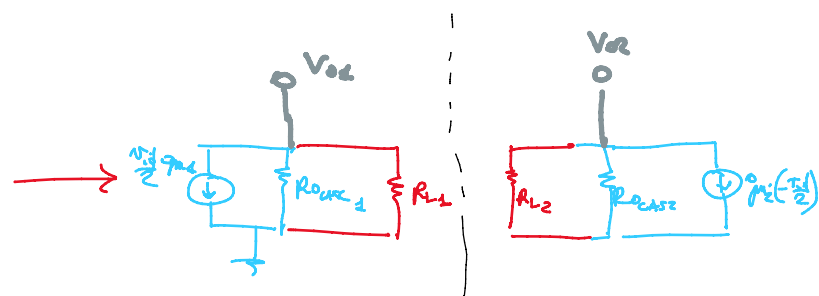
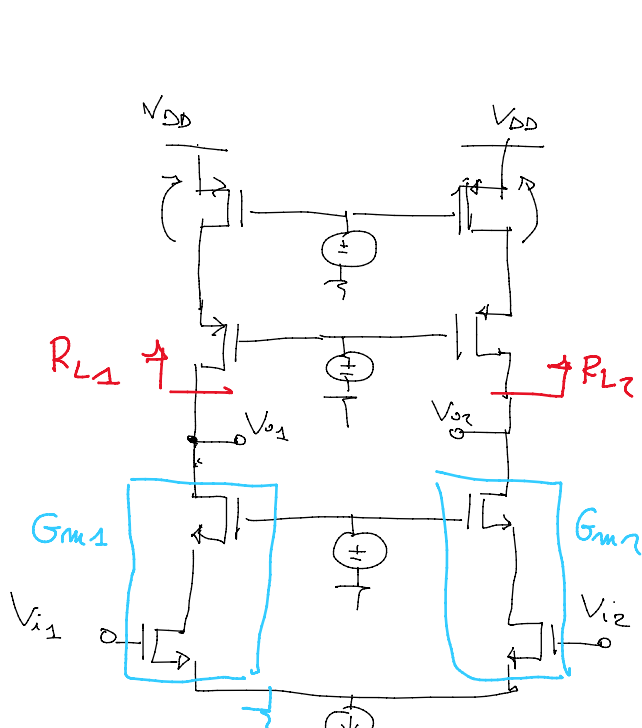
Cascode is high when compared to Common source stage.

class Note



$$G_{L1} = 0$$

$$\frac{v_{od}}{v_{id}} = -g_m \cdot R_{oCASC}$$

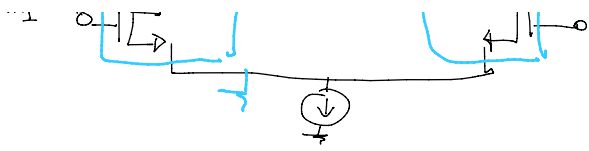


$$v_{o1} = \frac{v_{id}}{2} \cdot g_{m1} (R_{oCASC1} \parallel R_{L1}) \quad v_{o2} = \frac{v_{id}}{2} (g_{m2} (R_{L2} \parallel R_{oCASC2}))$$

$$\frac{v_{od}}{v_{id}} = -[g_m (R_{oCASC} \parallel R_L)]$$

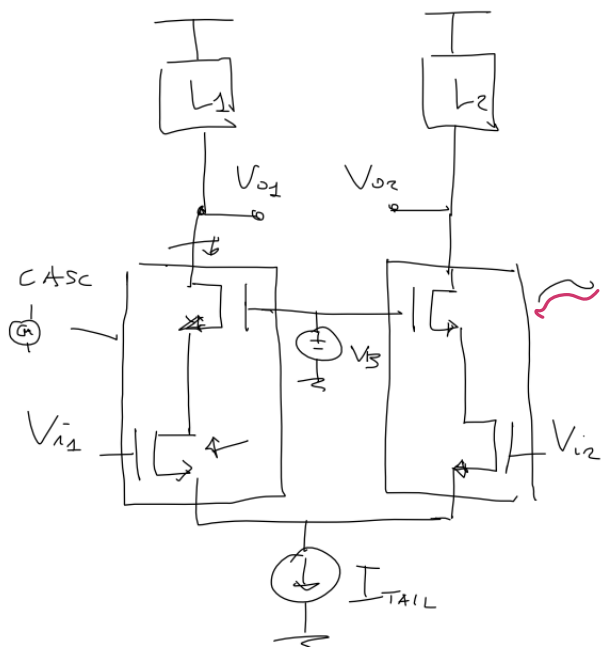
$$R_L = R_{out_CASC\ p}$$

$$\frac{v_{od}}{v_{id}} = -g_m (R_{oCASC\ n} \parallel R_{out_CASC\ p})$$



$$\frac{v_{od}}{v_{id}} = -g_m (R_{O_{cascN}} \parallel R_{O_{cascP}})$$

Differential pair Based on Cascode topology



if

$$G_{L1} = 0$$

$$\frac{v_{od}}{v_{id}} = -g_m \cdot R_{O_{casc}}$$

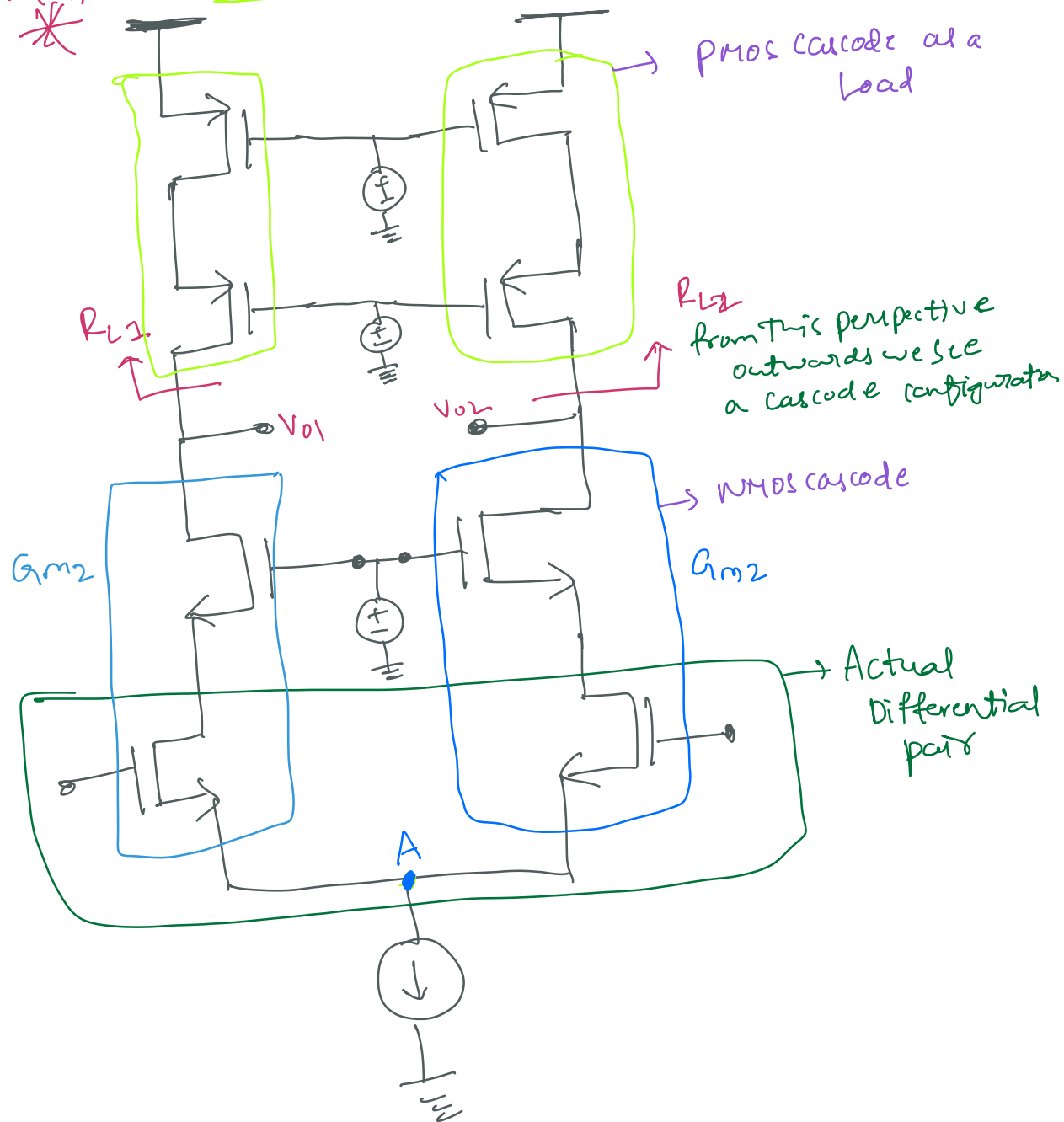
differential gain

Here G_m is a Cascode Amplifier

→ From the General abstraction of the Differential we can choose what ever amplifier we want for G_m and we can choose what ever load we want for L depending on the Requirement.

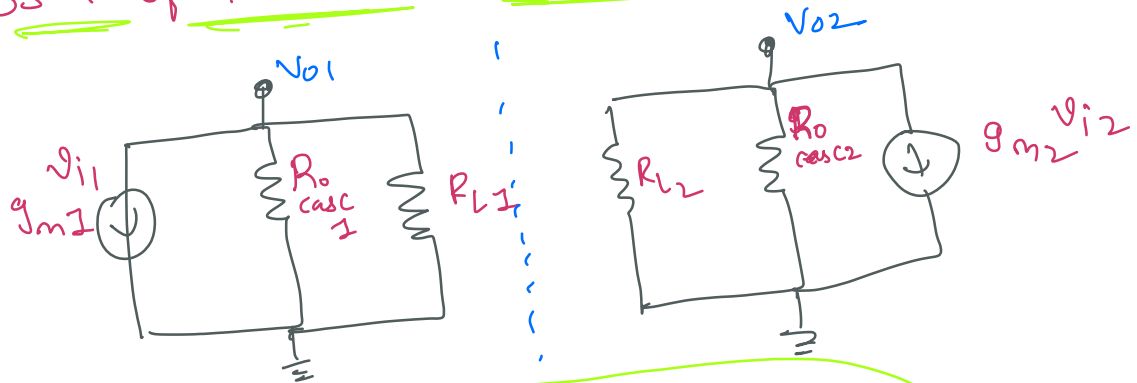
Q What if choose Cascode stage for load L and Cascode stage for Amplifier G_m ?

*** Telescopic Differential Amplifier



→ Node A will be a virtual ground when we are considering a "Differential Gain".

SSM of the above configuration.



$$V_c = \frac{v_{i1} + v_{i2}}{2}, \quad V_d = v_{i1} - v_{i2}$$

$$\Rightarrow \textcircled{v_{i1}} = V_c + \frac{V_d}{2}$$

$$\textcircled{v_{i2}} = V_c - \frac{V_d}{2}$$

$$v_{i1} = v_{ic} + \frac{v_{id}}{2}$$

$$v_{i2} = v_{ic} - \frac{v_{id}}{2}$$

for differential Gain, $V_c = 0$

$$\therefore \textcircled{v_{i1} = \frac{v_{id}}{2}}, \quad \textcircled{v_{i2} = -\frac{v_{id}}{2}}$$

Main Calculation

$$v_{o1} = -\frac{v_{id}}{2} \cdot g_{m1} (R_{o_{casc1}} \parallel R_{L2}) \quad \text{--- (1)}$$

$$v_{o2} = \frac{v_{id}}{2} g_{m2} (R_{L2} \parallel R_{o_{casc2}}) \quad \text{--- (2)}$$

from (1) & (2)

$$\frac{V_{od}}{V_{id}} = -g_m (R_{o_{casc}} \parallel R_L)$$

$$\text{if } \begin{cases} R_{L1} = R_{L2} \\ R_{o_{casc1}} = R_{o_{casc2}} \end{cases}$$

$$R_L = R_{out_{cascp}}$$

$$\frac{V_{od}}{V_{id}} = -g_m (R_{o_{casc}} \parallel R_{out_{cascp}})$$

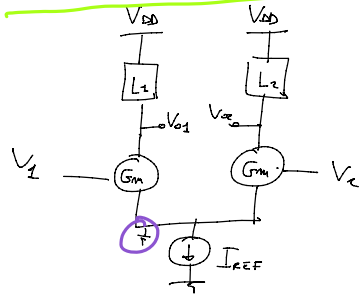
→ The Telescopic Differential Amplifier offers a High Gain.

But can we use it in all cases, the Answer would be NO.

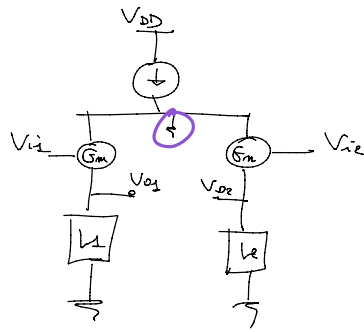
Depending on the functionality we would choose the type of Differential Amplifier we want.

**

Folded Cascode version of Differential pair.



NMOS Differential pair

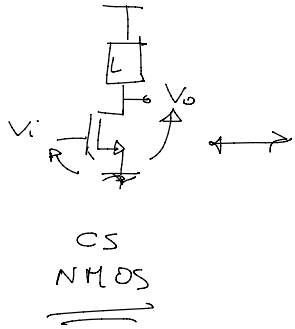


PMOS Differential pair

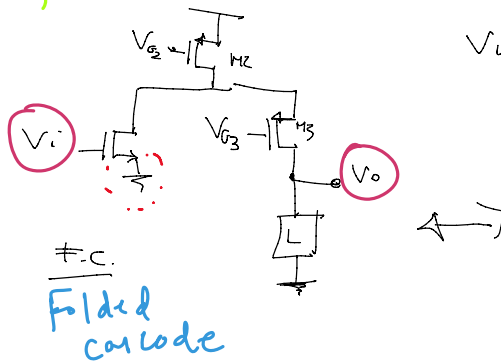
$\frac{1}{\infty} \rightarrow$ virtual GND

G_m can be implemented by any scheme

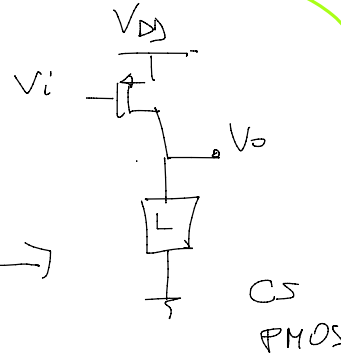
$$G_m \equiv \frac{I}{V_{GS}}$$



CS NMOS

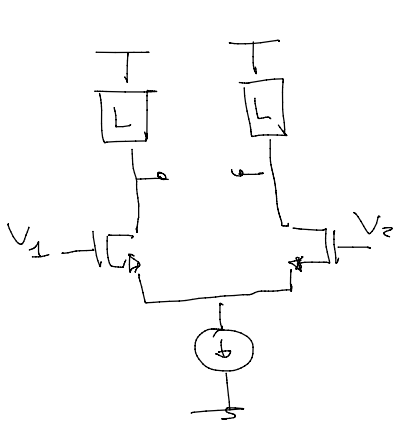


$\pm$ -c. Folded cascode

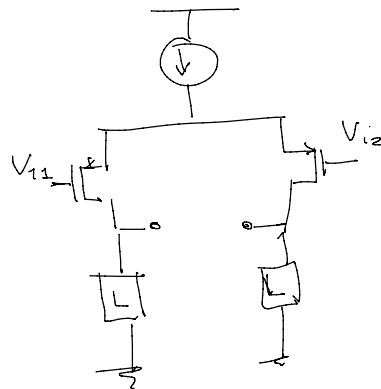


CS PMOS

Here we have 2 types of G_m

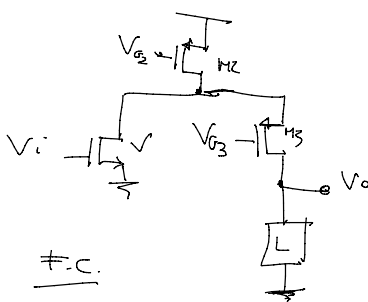


NMOS Differential pair

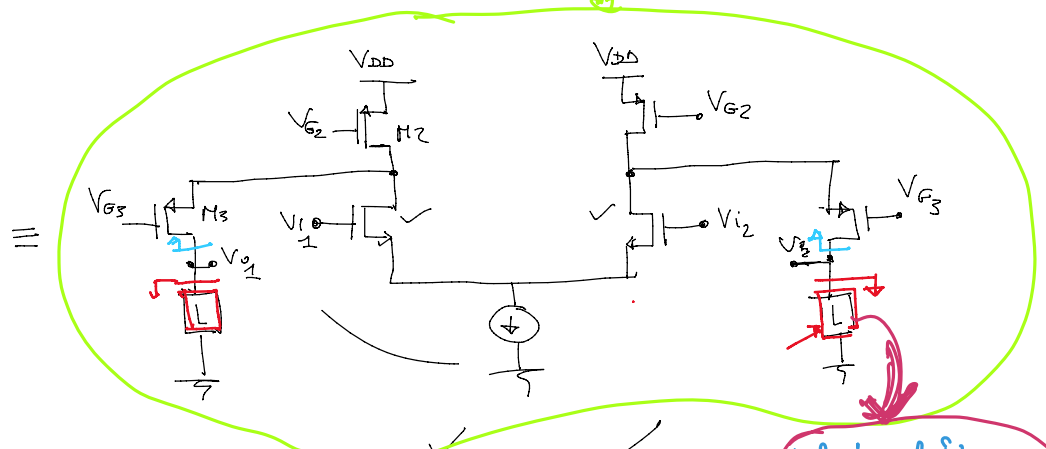


PMOS Differential pair

Implementation of "Differential pair using Folded Cascode"



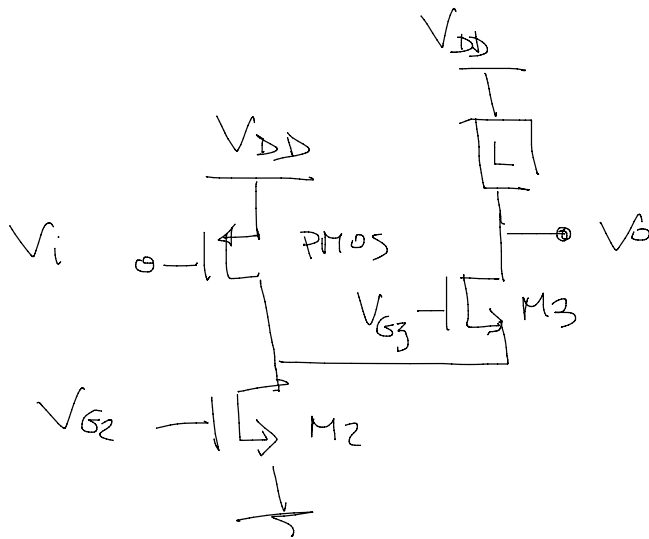
$\pm$ -c.



$$V_{od} = -g_m \cdot (R_{out,FC} \parallel R_{out,CASC})$$

if load is a cascode load.

$$\frac{V_{od}}{V_{id}} = -g_m \cdot (\overset{\checkmark}{R_{out_{FC}}} \parallel \overset{\checkmark}{R_{out_{casc}}})$$



$\Rightarrow$ Diff. Pair?

By

Revanth Reddy Pannala

EBIT, Unibo